Linear Combination and Coplanar Vectors

Recall: Two vectors $\vec{u}$ and $\vec{v}$ are collinear if $\vec{u} = k\vec{v}$, $k \in \Re$. (i.e. They are scalar multiples).

Linear Combination in R^2

Linear combination of vectors is when we express _____ vector as a ____ of two other vectors.

i.e. $\overrightarrow{OP} = (3,2)$ can be expressed as $\overrightarrow{OP} =$

Every vector in R^2 can be written as a linear combination of $\hat{i}$ and $\hat{j}$, therefore $\hat{i}$ and $\hat{j}$ are said to _____ for R^2. In fact, they are called the _____ vectors for R^2.

In fact, as long as two vectors $\vec{u}$ and $\vec{v}$ are _____, they can also form a basis in R^2.

Example 1: Express $\vec{w} = (6,8)$ as a linear combination of $\vec{u} = (3,5)$ and $\vec{v} = (1,2)$.

Coplanar Vectors in R^3

Vectors are said to be _____ if they lie in the same plane. (i.e. one can be expressed as a linear combination of the other two).

Three vectors, $\vec{u}$, $\vec{v}$, and $\vec{w}$ are coplanar if and only if there exists two scalars *a* and *b* such that $\vec{w} = a\vec{u} + b\vec{v}$.

Example 2: Determine if $\vec{u} = (3,-1,4)$, $\vec{v} = (6,-4,-8)$, and $\vec{w} = (7,-3,4)$ are coplanar?

Is there a quicker way to determine if 3 vectors are coplanar?

Triple Scalar Product $\vec{u} \bullet (\vec{v} \times \vec{w})$

If $\vec{u} \bullet (\vec{v} \times \vec{w}) = 0$, then the non-zero vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$ are coplanar.

Example 3: Determine if $\vec{u} = (3,-1,4)$, $\vec{v} = (6,-4,-8)$, and $\vec{w} = (7,-3,4)$ are coplanar?